Lecture 14

# The Dirac field

A field equation can be first order in both time and space. The Dirac field provides such an example. We construct its evolution and recover a conserved current and a real symplectic form. Its conserved positive norm does not make its classical energy positive.

**Objectives.** *We derive and solve the two-component Dirac equation in one space dimension. We prove finite propagation speed and identify its symplectic form on the underlying real solution space.*

By the end of this lecture you will be able to:

- Derive the Dirac equation and its adjoint from the action.
- Construct the unique distributional evolution for arbitrary $L^2$ data.
- Compute mode currents and distinguish positive charge from energy.

## 14.1 Matrices, field and action

**Definition 14.1** (Clifford relation and Dirac matrices)**.** With $\eta = \operatorname{diag}(1,-1)$, the *Clifford relation* is $\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2\eta^{\mu\nu} I$ for $\mu,\nu = 0,1$. Our *Dirac matrices* are

$$\gamma^0 = \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \gamma^1 = \mathrm{i}\sigma_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Here $I$ is the two-dimensional identity matrix.

Their squares are $I$ and $-I$, and their mixed anticommutator is zero. Define the Hermitian matrices

$$\alpha = \gamma^0\gamma^1 = -\sigma_3 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \beta = \gamma^0.$$

They obey $\alpha^2 = \beta^2 = I$ and $\alpha\beta + \beta\alpha = 0$.

**Definition 14.2** (Spinor field and Dirac adjoint)**.** A *spinor field* here is a two-component column $\psi : \mathbb{R}^{1+1} \to \mathbb{C}^2$. Its entries are ordinary commuting complex numbers. Its *Dirac adjoint* is the row $\bar\psi = \psi^\dagger\gamma^0$, where $\dagger$ denotes conjugate transpose.

For smooth fields and $m \geq 0$, the Dirac density is

$$L = \bar\psi(\mathrm{i}\gamma^\mu\partial_\mu - m)\psi = \mathrm{i}\psi^\dagger\psi_t + \mathrm{i}\psi^\dagger\alpha\psi_x - m\psi^\dagger\beta\psi. \tag{14.1}$$

During variation, treat $\psi$ and $\psi^\dagger$ as independent complex variables and then restore conjugacy. On a compact spacetime rectangle, integrating the derivatives of $\delta\psi$ gives the boundary term $\int_{\partial D} \mathrm{i}\bar\psi\gamma^\mu\delta\psi\,\mathrm{d}\Sigma_\mu$. For variations supported in the interior this term vanishes.

The remaining coefficients of $\delta\psi^\dagger$ and $\delta\psi$ are

$$\mathrm{i}\psi_t + \mathrm{i}\alpha\psi_x - m\beta\psi, \qquad -\mathrm{i}\psi_t^\dagger - \mathrm{i}\psi_x^\dagger\alpha - m\psi^\dagger\beta.$$

They are adjoints of each other. Stationarity gives

$$\mathrm{i}\psi_t = \mathsf{H}\psi, \qquad \mathsf{H} = -\mathrm{i}\alpha\partial_x + m\beta. \tag{14.2}$$

The operator $\mathsf{H}$ maps Schwartz columns to Schwartz columns. The density need not be pointwise real. The product rule gives $L - \overline{L} = \mathrm{i}\partial_t(\psi^\dagger\psi) + \mathrm{i}\partial_x(\psi^\dagger\alpha\psi)$. Its imaginary part contributes only a boundary term to the action. Thus compact variation agrees with variation of the real action in the real and imaginary components.

## 14.2 The Cauchy problem in $L^2$

For rough data, a solution means $\psi \in C(\mathbb{R}; L^2(\mathbb{R}, \mathbb{C}^2))$ satisfying (14.2) in distributions, componentwise against compact smooth spacetime tests, with the prescribed value at zero in the $L^2$ norm. The Fourier convention remains that of Lecture 4. The symbol and frequency are

$$H(k) = \alpha k + m\beta = \begin{pmatrix} -k & m \\ m & k \end{pmatrix}, \qquad H(k)^2 = \nu(k)^2 I, \qquad \nu(k) = \sqrt{k^2 + m^2}. \tag{14.3}$$

**Theorem 14.3** (Dirac evolution). *For every $f \in L^2(\mathbb{R}, \mathbb{C}^2)$ and $m \geq 0$, there is exactly one solution in the stated class, given by*

$$\widehat{\psi}(t, k) = U(t, k)\widehat{f}(k), \qquad U(t, k) = \cos(t\nu)I - \mathrm{i}S_t(\nu)H(k). \tag{14.4}$$

*Here $S_t(\nu) = \sin(t\nu)/\nu$ for $\nu > 0$ and $S_t(0) = t$. The evolution is unitary: $\|\psi(t)\|_2 = \|f\|_2$. Both components solve Klein–Gordon distributionally. If $f$ vanishes outside a closed set $K$, then $\psi(t)$ vanishes almost everywhere outside $\{x : \operatorname{dist}(x, K) \leq |t|\}$.*

*Proof.* Hermiticity of $H(k)$ and its square give $U^\dagger U = (\cos^2(t\nu) + \nu^2 S_t(\nu)^2)I = I$. Also $\mathrm{i}U_t = H(k)U$ and $U(0, k) = I$. At $m = k = 0$, the symbol vanishes and $U = I$. Pointwise continuity and the bound $\|U\widehat{f}\| \leq \|\widehat{f}\|$ permit dominated convergence in $L^2$. Plancherel proves time continuity and norm preservation. Applying the same argument to a difference of data proves continuous dependence, uniformly for all times.

Multiply $\widehat{f}$ by a smooth compact frequency cutoff equal to one on $[-R, R]$, as in Lecture 4. The inverse transform is a smooth solution: all differentiated inverse integrals converge absolutely on compact time intervals. Unitarity makes these solutions converge to $\psi$ uniformly in time in $L^2$. Passing to the limit against compact tests proves the distributional equation.

For uniqueness, cut off the Fourier transform of a difference with zero data. The passage from compact spatial tests to Schwartz tests uses the $L^2$ cutoff argument of Lecture 4 and justifies this transformed equation. On its compact frequency support, $H(k)$ is a bounded Hermitian operator on $L^2$. Its distributional equation therefore gives a continuous $L^2$ derivative: integrate $-\mathrm{i}Hw$ in the norm and use the scalar zero-derivative argument of Lecture 4 to identify $w$ with that primitive. Thus $w$ is $C^1$ in $L^2$, and $\frac{\mathrm{d}}{\mathrm{d}t}\|w\|_2^2 = 2\operatorname{Re}(w, -\mathrm{i}Hw) = 0$. The zero datum forces $w = 0$. Removing the cutoff gives uniqueness.

Constant-coefficient distributional derivatives commute. Apply $\mathrm{i}\partial_t + \mathsf{H}$ to $(\mathrm{i}\partial_t - \mathsf{H})\psi = 0$. Since $\mathsf{H}^2 = -\partial_x^2 + m^2$, this gives $(\partial_t^2 - \partial_x^2 + m^2)\psi = 0$. For arbitrary $L^2$ data, the initial

velocity specified by this equation is the distribution $-\mathrm{i}\mathsf{H}f$, which need not belong to $L^2$. This componentwise assertion is therefore not the energy-class Klein–Gordon statement of Lecture 4.

For propagation, start with smooth compact data. Their solution is smooth by the Fourier formula. Define its charge density and current by

$$\rho_D = j^0 = \psi^\dagger\psi, \qquad j^1 = \bar\psi\gamma^1\psi = \psi^\dagger\alpha\psi, \qquad |j^1| \le \rho_D. \tag{14.5}$$

The inequality follows from $\rho_D + j^1 = 2|\psi_2|^2$ and $\rho_D - j^1 = 2|\psi_1|^2$. Using the Dirac and adjoint equations gives

$$\partial_t\rho_D = -\psi_x^\dagger\alpha\psi - \psi^\dagger\alpha\psi_x = -\partial_x j^1.$$

The mass terms cancel. Coordinate FTC on $I_t = [a - R + t, a + R - t]$, for $0 \le t < R$, yields

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{I_t}\rho_D\,\mathrm{d}x = -(\rho_D + j^1)(t, a + R - t) - (\rho_D - j^1)(t, a - R + t) \le 0.$$

This is the moving-interval calculation of Lecture 5, now with a density that controls the field itself. Zero initial charge on the base interval forces the solution to vanish on every smaller interval. Approximate arbitrary $L^2$ data by compact smooth data. Unitarity gives $L^2$ convergence at the initial and final times, so the local squared norms and their inequality converge. The support conclusion follows by choosing a base interval disjoint from $K$, as in Lecture 5. Time reversal handles $t < 0$. ∎

The conserved total charge is $Q_D = \int\rho_D\,\mathrm{d}x = \|f\|_2^2$. For rough solutions the currents are $L^1$ functions, and their continuity identity follows by the same $L^2$ approximation. The construction never requires pointwise values of an arbitrary $L^2$ spinor.

## 14.3 Modes and the sign of energy

A plane-wave mode has the form $\psi(t,x) = ue^{-\mathrm{i}\lambda t + \mathrm{i}kx}$, where $u \in \mathbb{C}^2$ and $H(k)u = \lambda u$. For $\nu > 0$, the eigenvalues are $\lambda = \nu$ and $\lambda = -\nu$, since $H^2 = \nu^2 I$ and $\operatorname{tr} H = 0$. For a normalised eigenvector, the density and current are

$$\rho_D = 1, \qquad j^1 = \frac{k}{\lambda} = \pm\frac{k}{\nu}.$$

Indeed, $H\alpha + \alpha H = 2kI$, so its expectation in $u$ gives $2\lambda u^\dagger\alpha u = 2k$. For a concrete pair, take $k = 3$ and $m = 4$:

$$H(3) = \begin{pmatrix} -3 & 4 \\ 4 & 3 \end{pmatrix}, \qquad \nu(3) = 5, \qquad u_+ = \frac{1}{\sqrt5}\begin{pmatrix}1\\2\end{pmatrix}, \qquad u_- = \frac{1}{\sqrt5}\begin{pmatrix}-2\\1\end{pmatrix}.$$

Here $H(3)u_+ = 5u_+$ and $H(3)u_- = -5u_-$. The modes $u_+e^{-5\mathrm{i}t+3\mathrm{i}x}$ and $u_-e^{5\mathrm{i}t+3\mathrm{i}x}$ have currents $3/5$ and $-3/5$, respectively. These are densities of modes. A nonzero plane wave has infinite total charge on $\mathbb{R}$ and is not among the theorem's $L^2$ data.

For Schwartz data the Hamiltonian is the real functional

$$H[\psi] = \int_{\mathbb{R}}\psi^\dagger\mathsf{H}\psi\,\mathrm{d}x = \int_{\mathbb{R}}\widehat{\psi}^\dagger H(k)\widehat{\psi}\,\mathrm{d}k. \tag{14.6}$$

For Schwartz columns $\psi, h$, integration of $\int\psi^\dagger\mathsf{H}h\,\mathrm{d}x$ by parts has boundary term $-\mathrm{i}[\psi^\dagger\alpha h]_{-R}^{R}$ on $[-R, R]$. It vanishes for Schwartz columns, so $\mathsf{H}$ is symmetric on them. Its Hamiltonian

has both signs even on finite-charge data. Since $U(t,k)$ commutes with $H(k)$, its Fourier formula also proves conservation of this energy for Schwartz data. For $m > 0$, the smooth projectors $P_\pm(k) = \frac{1}{2}(I \pm H(k)/\nu)$ select the two eigenlines. Choose a nonzero smooth compact Fourier packet in either eigenline. Its inverse transform is Schwartz, and its energy is $\pm\int \nu|\widehat{\psi}|^2\,\mathrm{d}k$, while its charge is strictly positive. Positive charge therefore does not imply positive classical energy. The planned QFT course revisits this energy question together with quantisation and fermionic statistics.

## 14.4 The real symplectic form

The time boundary term of the action gives the momentum row $\pi_\psi = \mathrm{i}\psi^\dagger$ and the potential $\Theta = \mathrm{i}\int \psi^\dagger\delta\psi\,\mathrm{d}x$. With the sign of Lecture 8, its antisymmetric variation is

$$\Omega = -\delta\Theta = -\mathrm{i}\int \delta\psi^\dagger \wedge \delta\psi\,\mathrm{d}x = \int \delta\psi \wedge \delta\pi_\psi. \tag{14.7}$$

The components are summed. Write $\psi = a + \mathrm{i}b$, with real two-component columns $a$ and $b$. Direct expansion gives

$$\Theta = \int (b\cdot\delta a - a\cdot\delta b)\,\mathrm{d}x + \frac{\mathrm{i}}{2}\delta\|\psi\|_2^2,$$

$$\Omega = 2\sum_{r=1}^{2}\int \delta a_r \wedge \delta b_r\,\mathrm{d}x.$$

Thus the imaginary part of the potential is exact and its exterior derivative vanishes. The symplectic form is real on the underlying real $L^2$ space. Explicitly,

$$\Omega(h_1,h_2) = 2\,\mathrm{Im}\int h_1^\dagger h_2\,\mathrm{d}x, \qquad \Omega(h,\mathrm{i}h) = 2\|h\|_2^2.$$

It is closed because it is constant, and the last identity proves nondegeneracy. Unitary evolution preserves the complex inner product and hence this form, so it is independent of the constant time slice. On Schwartz paths, $X_H = -\mathrm{i}\mathsf{H}\psi$ satisfies $\iota_{X_H}\Omega = \delta H$ and is the Dirac evolution. No anticommuting variables or additional constraint algorithm were used.

## 14.5 The three answers in Part I

The constructions of Part I can now be compared on their stated domains.

| Field | Action | Cauchy problem | Canonical structure |
|---|---|---|---|
| Scalar | Compact variations give Klein–Gordon. | $H^1\times L^2$ data give unique causal evolution. | Slice-independent form on compact-data solutions. |
| Maxwell | Gauge invariance gives an identity. | Conserved compact source and compatible compact data in Lorenz gauge. | Vacuum annulus gauge quotient. |
| Dirac in 1 + 1 | Independent adjoint variation gives a first-order equation. | $L^2$ spinors give unique unitary causal evolution. | Real symplectic form on the complex $L^2$ solution space. |

The four-component Dirac equation in 3 + 1 dimensions is Lorentz covariant under transformations preserving spacetime orientation and the future direction. This statement is *cited*: see Thaller, *The Dirac Equation* (1992), §3.3.8, pp. 100–101, Chapter 3. Its spin transformations use the double cover $SL(2,\mathbb{C})$ of that Lorentz group, recalled in *Lie Groups*, Lecture 24, and *Symmetries*, Lecture 17.

**Summary.** In 1+1 dimensions the two-component Dirac equation has unique unitary causal evolution for every $L^2$ datum. Its components solve Klein–Gordon distributionally, while its positive charge and indefinite classical energy are distinct conserved quantities on their stated domains. The action induces a real, nondegenerate, slice-independent form on the underlying real solution space.

**Self-check.** At $m = 0$, solve the two component transport equations from their initial profiles. Which component carries current to the right?

**Answer.** The components are $\psi_1(t,x) = f_1(x+t)$ and $\psi_2(t,x) = f_2(x-t)$. The second component travels right with $j^1 = |\psi_2|^2$ when the first is zero.

**Self-check.** At $m = 0$, take $f = (\mathbf{1}_{[0,1]}, 0)^T$. Is the time derivative at zero an $L^2$ function? Does componentwise Klein–Gordon put this datum in L4's class?

**Answer.** The first time derivative is $\delta_0 - \delta_1$, a distribution rather than an $L^2$ function. The first component solves Klein–Gordon distributionally, but this is not an $H^1 \times L^2$ Cauchy datum.

**Self-check.** For the worked negative-energy mode, compare the charge density, energy density and total charge. Can finite-charge states also have negative energy?

**Answer.** The densities are 1 and −5. The plane wave has infinite total charge. A nonzero smooth Fourier packet in the negative eigenline has finite positive charge and strictly negative energy.

**Self-check.** If a real tangent vector $h$ pairs to zero with every tangent vector under $\Omega$, which test vector proves $h = 0$?

**Answer.** Choose i$h$, viewed as another vector in the underlying real space. The pairing is $2\|h\|_2^2$, so $h = 0$.